

WeatherSphere Pro — QA Test Cases Report

Test Case ID	Test Scenario	Test Steps
TC-001	Valid city search	Enter a valid city in the search box and click Search
TC-002	Search for another valid city	Enter a valid city in the search box and click Search
TC-003	Search for an invalid city	Enter an invalid city in the search box and click Search
TC-004	Search with empty input	Leave the search box empty and click Search
TC-005	Search with leading and trailing spaces	Enter a valid city name with spaces before and after it, then click Search
TC-006	Search using different letter case	Enter a valid city name using mixed uppercase and lowercase letters, then click Search
TC-007	Search with a very long city name	Enter a very long city name in the search box and click Search
TC-008	Search button functionality	Enter a valid city name and click the Search button
TC-009	Add a city to favorites	Search for a city and click the Favorite button
TC-010	Verify favorite city persists after refresh	Add a city to favorites, refresh the page, and check the favorites list
TC-011	Verify multiple favorite cities persist after refresh	Add multiple cities to Favorites, refresh the page, and check the Favorites list
TC-012	Open weather for a favorite city	Click a saved city from the Favorites section
TC-013	Verify temperature remains consistent after refresh	Note the displayed temperature, refresh the page, and compare the temperature after loading
TC-014	Verify humidity value against API	1. Open WeatherSphere for Mumbai 2. Note humidity 3. Check the corresponding Open-Meteo API value 4. Refresh WeatherSphere and compare again
TC-015	Verify city search with lowercase input	Enter a city name using lowercase letters and click Search
TC-016	Verify city search with uppercase input	Enter a city name using uppercase letters and click Search
TC-017	Verify city search with special characters and invalid input	Enter a city name containing special characters and numbers, then click Search

TC-018	Verify behavior when search field is empty	Leave the city search field empty and click Search
TC-019	Verify search for a multi-word city name	Enter a valid city name containing multiple words and click Search
TC-020	Verify behavior for a non-existent city	Enter a properly formatted but non-existent city name and click Search
TC-021	Verify wind speed display against API value	1. Open WeatherSphere for New Delhi 2. Note the displayed wind speed 3. Check the Open-Meteo current wind speed 4. Compare the values considering rounding
TC-022	Verify humidity value display format	Open the weather information for New Delhi and check the displayed humidity value
TC-023	Verify temperature unit display	Open the weather information for New Delhi and check the main temperature display
TC-024	Verify 7-day weather forecast is displayed	Open the Forecast section and count the displayed forecast day entries
TC-025	Verify forecast weekday sequence	Check the weekday labels displayed in the 7-day forecast from the first entry to the last
TC-026	Verify high and low temperatures are displayed for all forecast days	Open the 7-day forecast and check each forecast card for both high and low temperature values
TC-027	Validate high and low temperature relationship	Compare the high and low temperatures displayed for each of the 7 forecast days
TC-028	Verify duplicate recent searches are not created	Search for the same city twice and check the Recent Searches section
TC-029	Verify weather insights update when city changes	1. Select New Delhi 2. Observe Weather Insights 3. Search Mumbai 4. Observe the same Weather Insights section again
TC-030	Verify Weather Insights remain available after page refresh	1. Select Mumbai 2. Open Weather Insights 3. Refresh the page 4. Return to Weather Insights

Test Data	Expected Result	Actual Result
Delhi	Weather information for Delhi should be displayed	Weather information for Delhi was displayed
Mumbai	Weather information for Mumbai should be displayed	Weather information for Delhi was displayed
asdfghjkl	An appropriate error message should be displayed indicating that the city could not be found	Unable to find city
No input	Application should provide appropriate feedback when the search is submitted without a city name	Nothing happened
Mumbai	Application should recognize the valid city name despite leading and trailing spaces	Mumbai weather information was displayed
mUmBal	Application should recognize the city name regardless of letter case	Mumbai weather information was displayed
DelhiDelhiDelhiDelhiD elhiDelhiDelhiDelhiDel hiDelhiDelhiDelhiDelhi	Application should handle long input without breaking the user interface	Unable to find city was displayed and the search box layout remained normal
Delhi	Search button should process the entered city and display its weather information	Delhi weather information was displayed
Mumbai	Selected city should be added to the favorites list	Mumbai was added to the favorites list
Mumbai	Favorite city should remain available after page refresh	Mumbai remained in the favorites list after refresh
Mumbai, Delhi	All added favorite cities should remain available after page refresh	Mumbai and Delhi remained in the Favorites list after refresh
Mumbai	Weather information for the selected favorite city should be displayed	Mumbai weather information was displayed after selecting Mumbai from Favorites
Mumbai	Temperature should be displayed correctly after page refresh without an unexpected or unrelated value	The same temperature was displayed after refresh
Mumbai	Dashboard humidity should match the current humidity value returned by the configured API when checked at the same time	After refresh, WeatherSphere displayed 77% and the API returned 77%
mumbai	The application should recognize the city and display its weather information	Mumbai weather was displayed correctly
MUMBAI	The application should recognize the city regardless of letter case and display its weather information	Mumbai weather was displayed correctly
Mumbai@123	Application should reject the invalid city input and display an appropriate error/message	Application displayed "Unable to find city"

Empty input	Application should display a clear validation message asking the user to enter a city name	Search button responds to the click, but no message or result is displayed
New Delhi	Application should recognize the multi-word city name and display its weather information	New Delhi weather information was displayed correctly
AaaaaaCity	Application should reject the city and display an appropriate error message	Application displayed "Unable to find city"
New Delhi	Displayed wind speed should correspond to the API value, accounting for the dashboard's rounding	API returned 1.3 km/h and WeatherSphere displayed 2 km/h, consistent with rounding
New Delhi	Humidity should be displayed as a percentage with a clear % symbol	Humidity was displayed as 75%
New Delhi	Temperature should be displayed with the Celsius (°C) unit	Temperature was displayed as 29°C
New Delhi	The application should display forecast information for 7 days	Forecast information for 7 days was displayed
New Delhi	Forecast weekday labels should appear in chronological weekday order	Weekday labels were displayed sequentially from Monday to Sunday
New Delhi	Each forecast day should display both high and low temperature values	All 7 forecast cards displayed both high and low temperature values
New Delhi	For every forecast day, the high temperature should be greater than or equal to the low temperature	High temperature was greater than or equal to low temperature for all 7 forecast days
Delhi	The same city should not be unnecessarily duplicated in the Recent Searches list	Delhi appeared only once after being searched twice
New Delhi → Mumbai	Weather Insights data should update according to the newly selected city	Weather Insights data changed after switching from New Delhi to Mumbai
Mumbai	Weather Insights charts should load normally after page refresh	Weather Insights chart appeared normally with data after refresh

Status

PASS

PASS

PASS

NEEDS REVIEW

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

FAIL

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

PASS

WeatherSphere Pro — QA Bug Report

Bug ID	Bug Title	Steps to Reproduce	Test Data
BUG-001	Favorite heart does not respond to click	1. Search Mumbai 2. Add Mumbai to Favorites 3. Click the heart icon on the Mumbai favorite card	Mumbai, Delhi
BUG-002	Light mode displays inconsistent colors and poor text visibility	1. Open WeatherSphere 2. Click the moon/theme button 3. Observe the page	Theme: Dark Mode → Light Mode
BUG-003	No validation message displayed for empty city search	1. Open WeatherSphere 2. Leave the city search field empty 3. Click Search	Empty input

Expected Result	Actual Result	Status	Severity
Clicking the heart should perform the available favorite/unfavorite action	Clicking the favorite heart does not remove the selected city. The issue occurs with both Mumbai and Delhi, and the cities remain in Favorites after refresh.	OPEN	Medium
Entire website should switch consistently to light mode with readable text and properly styled backgrounds/cards.	After switching to Light Mode, some elements remain dark while others become light, causing poor text visibility. After refreshing the page, Dark Mode returns. The issue is reproducible when switching to Light Mode again.	OPEN	Medium
The application should display a clear validation message asking the user to enter a city name.	The Search button responds to the click, but no message or result is displayed.	OPEN	Low

Priority

Medium

High

Medium

WEATHERSPHERE PRO – TEST SUMMARY REPORT

Project Name	WeatherSphere Pro – Weather Dashboard
Testing Type	Manual Functional & UI Testing
Testing Scope	City Search, Favorites, Weather Data, Forecast, Recent Searches, Weather Insights
Total Test Cases	30
Passed	28
Failed	1
Needs Review	1
Pass Percentage	93.33%
Confirmed Defects	3
Testing Status	Completed

TEST EXECUTION SUMMARY

Result	Count	Percentage
PASS	28	93.33%
FAIL	1	3.33%
NEEDS REVIEW	1	3.33%

DEFECT SUMMARY

Bug ID	Description	Severity	Priority
BUG-001	Favorite heart does not respond to click	Medium	Medium
BUG-002	Light mode visibility/readability issue	Medium	High
BUG-003	Empty search input validation issue	Low	Medium

**QA
CONCLUSION**

A total of 30 test cases were executed for the WeatherSphere Pro weather dashboard. 28 test cases passed, 1 test case failed and 1 test case requires review, resulting in a 93.33% pass rate. A total of 3 defects were identified and documented with severity and priority. The current testing cycle has been completed.

Test Execution Status

